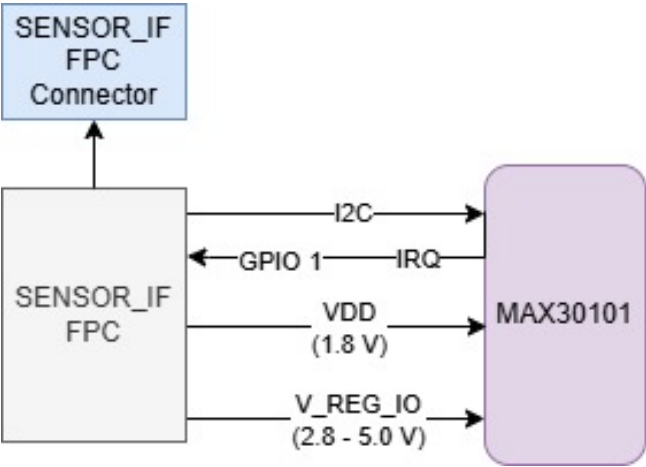


Schematic Sheets

#	Name	File	Description
1	Main	SensWear_PPG	Optical AFE IC and connectors
2			
3			
4			

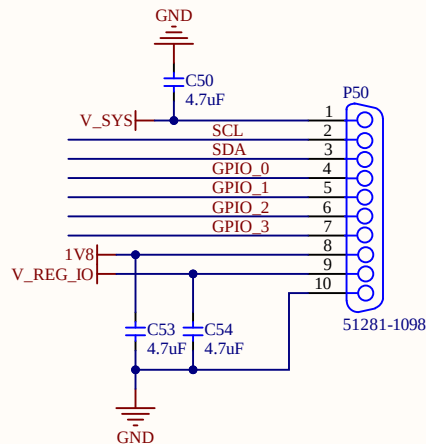
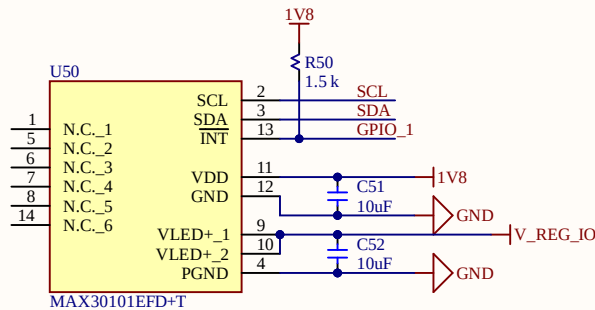
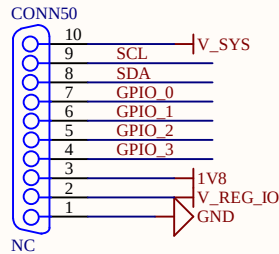
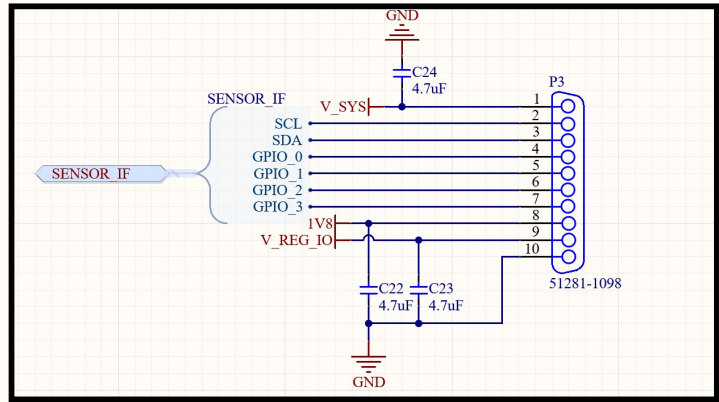
Revision History

Revision	Date	Description
A0	May 2025	Initial validation design on rigid PCB
B0	March 2026	Flexible PCB manufacturing updates



SensWear-V1R1

Designer/Engineer	Board Title		
Yusein Ali	SensWear-PPG		
Drawn By	Document Name		
Yusein Ali	SensWear-V1R1-PPG-		
Approved By	Ver.	Rev.	Date
Dariush Salami	1	B0	01.03.2026



PPG SENSOR (MAX30101) USAGE NOTES

Logic Level : 1.8V VDD
LED Supply : V_REG_IO / VLED+

[1. INTERFACE]

Protocol : I2C

Speed : Up to 400 kHz

Address : I2C21, 0x57, 7-bit Zephyr address
0xAE write / 0xAF read

[2. PIN MAPPING]

Signal	MAX30101 Pin	Net
SCL	Pin 2	SYS_I2C_SCL (P1.03)
SDA	Pin 3	SYS_I2C_SDA (P1.02)
INT#	Pin 13	SENSOR_GPIO_1 (P10.14)
VDD	Pin 11	1V8
VLED+	Pins 9,10	V_REG_IO
GND	Pin 12	GND
PGND	Pin 4	GND

[3. NOTES]

INT# is active-low open-drain.

SCL/SDA and INT# need pull-ups to 1V8.

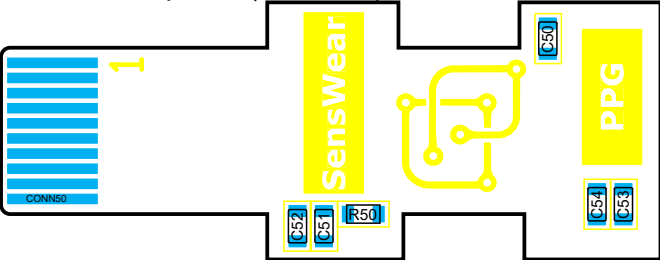
N.C. pins: 1, 5, 6, 7, 8, 14.



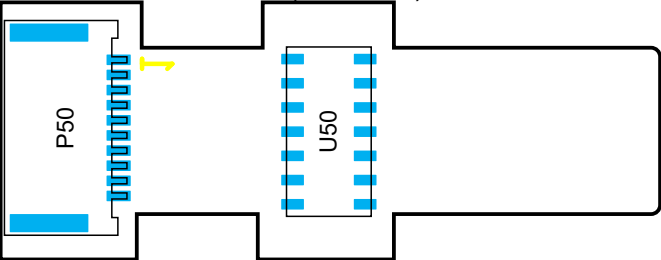
SensWear-V1R1

Designer/Engineer Yusein Ali	Board Title SensWear-PPG
Drawn By Yusein Ali	Document Name SensWear_PPG.SchDoc
Approved By Dariush Salami	Rev. B0 Sheet 1 of 1 Date 01.03.2026 Size A4

View from Top side (Scale 4:1)



View from Bottom side (Scale 4:1)



Bill Of Materials

Line #	Designator	Comment	Quantity
1	C50, C53, C54	4.7uF	3
2	C51, C52	10uF	2
3	CONN50	NC	1
4	P50	51281-1098	1
5	R50	1.5 k	1
6	U50	MAX30101EFD+T	1